counting squares gives total area = (4.0 ± 2.0) × 10 J kg B1

3 **(a)** **(i)** g and r are constant, so a is proportional to x B1

negative sign shows a and x are in opposite direction B1

(ii) $\omega^2 = \frac{g}{r}$ and $\omega = \frac{2\pi}{T}$ C1

$\omega^2 = \frac{9.81}{0.28} = 35$

$T = 1.06$ s M1

$\tau = 0.53$ s A1

(b) Sketch: time period constant (or increases very slightly) B1

drawn lines always ‘inside’ given loops, up to given time duration	B1
successive decrease in peak height	B1

4 (a) (i) $\tan\theta = \frac{38}{}$